DGMS JSO GEOLOGY 2019 • MINERALOGY & PETROLOGY

Question 95: Mineral Stoichiometry & Silica Saturation

Detailed Chemical Derivation, Binary Phase Equilibria, and Mutual Incompatibility of Nepheline and Quartz

Q.95: The chemical formula of which one of the following minerals can be expressed as a combination of nepheline and quartz?

(a) Anorthoclase

(b) Analcime

(c) Antigorite

✓ **(d) Albite**

Correct Option: (d) Albite

1. Stoichiometric Derivation

In mineralogy and petrology, silica saturation defines whether a rock or mineral contains sufficient SiO_2 to form fully saturated framework silicates (feldspars) or deficient SiO_2 requiring feldspathoids:

- **Nepheline (Feldspathoid):** An undersaturated sodium tectosilicate with the formula **$NaAlSiO_4$**.
- **Quartz (Silica):** Pure framework silica with the formula **SiO_2**.

- **Albite (Alkali Feldspar):** Saturated sodium tectosilicate with the formula **$NaAlSi_3O_8$**.

Stoichiometric Combination:

$$NaAlSiO_4 \text{ (Nepheline)} + 2\ SiO_2 \text{ (Quartz)} \rightarrow NaAlSi_3O_8 \text{ (Albite)}$$

- Sodium (Na): 1 = 1
- Aluminium (Al): 1 = 1
- Silicon (Si): 1 + 2 = 3
- Oxygen (O): 4 + (2 × 2) = 8

The sum of one formula unit of nepheline and two formula units of quartz yields exactly one formula unit of **albite**.

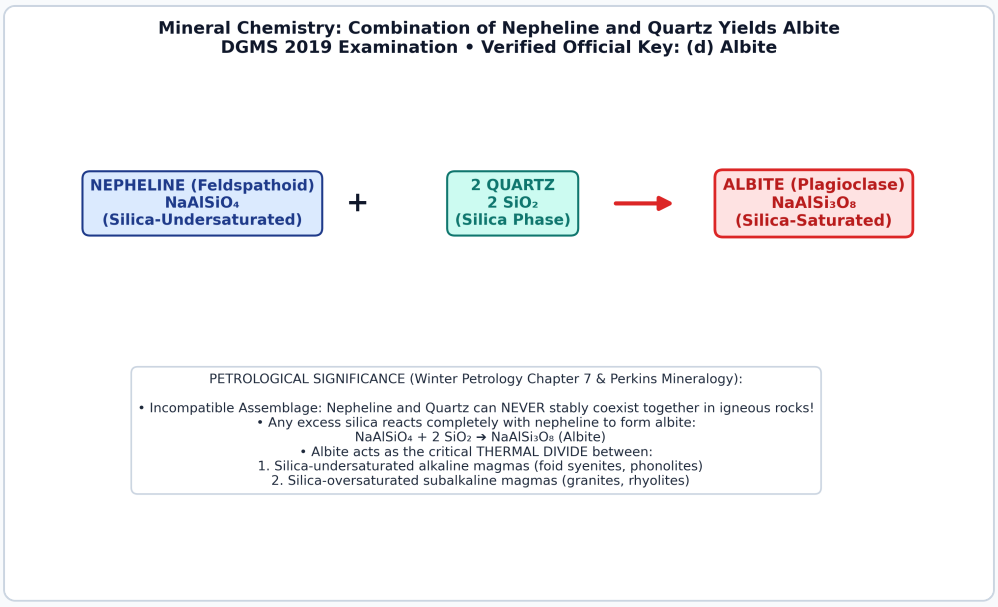


Figure 95.1: The Nepheline–Silica ($NaAlSiO_4$–SiO_2) binary system showing Albite ($NaAlSi_3O_8$) as a congruently melting thermal divide at 1118°C, separating the silica-undersaturated nepheline syenite field from the silica-oversaturated granite field. Based on Winter, Principles of Igneous and Metamorphic Petrology (Chapter 7).

2. Petrological Significance: The Thermal Divide

In the binary system $NaAlSiO_4$–SiO_2 (Bowen, 1937; Schairer & Yoder, 1960), albite melts congruently at 1118°C at atmospheric pressure. Albite forms a high-temperature **thermal divide** (or thermal barrier):

- Magmas with bulk composition to the right of albite (excess SiO_2) fractionate towards the quartz-albite eutectic (e.g., granites, rhyolites).
- Magmas with bulk composition to the left of albite (deficient in SiO_2) fractionate towards the nepheline-albite eutectic (e.g., nepheline syenites, phonolites).
- Because the free energy of reaction ΔG for *Nepheline + 2 Quartz → Albite* is strongly negative under crustal conditions, **nepheline and quartz can never coexist in thermodynamic equilibrium** in igneous or metamorphic rocks. Any excess silica immediately reacts with nepheline to synthesize albite.

3. Systematic Option Evaluation

Option	Mineral Name	Chemical Formula	Evaluation & Relationship to Nepheline + Quartz
(a)	Anorthoclase	$(Na,K)AlSi_3O_8$	**Incorrect.** Contains essential potassium (solid solution between albite and sanidine, typically $Or_{10}Ab_{90}$ to $Or_{36}Ab_{64}$); cannot be formed purely from pure sodium nepheline and quartz.
(b)	Analcime	$NaAlSi_2O_6 \cdot H_2O$	**Incorrect.** A hydrous zeolite/ feldspathoid. Stoichiometrically represents Nepheline + Quartz + H_2O, but the anhydrous combination of nepheline and quartz yields albite.
(c)	Antigorite	$Mg_3Si_2O_5(OH)_4$	**Incorrect.** A magnesium-rich serpentine-group phyllosilicate found in

Option	Mineral Name	Chemical Formula	Evaluation & Relationship to Nepheline + Quartz
			hydrated ultramafic rocks; contains Mg and OH, completely devoid of Na and Al.
(d)	**Albite**	**$NaAlSi_3O_8$**	**Correct.** Pure anhydrous framework silicate formed by combining 1 mole of nepheline ($NaAlSiO_4$) with 2 moles of quartz (SiO_2).

> **Key Petrological Takeaway**
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> **Feldspathoids** like nepheline ($NaAlSiO_4$) and leucite ($KAlSi_2O_6$) are silica-undersaturated equivalents of the alkali feldspars albite ($NaAlSi_3O_8$) and K-feldspar ($KAlSi_3O_8$). The presence of modal quartz immediately precludes the presence of modal nepheline in any equilibrium assemblage.

4. Standard Textbook References

- **John D. Winter**, *Principles of Igneous and Metamorphic Petrology* (2nd Edition, 2014), Chapter 7: "Binary Phase Systems", Section 7.4: Binary Systems with Congruent Intermediate Compounds ($NaAlSiO_4$–SiO_2), pp. 110–115.
- **W. A. Deer, R. A. Howie, & J. Zussman**, *An Introduction to the Rock-Forming Minerals* (3rd Edition, 2013), "Feldspathoids: Nepheline Group", pp. 412–425.
- **Dexter Perkins**, *Mineralogy* (3rd Edition, Pearson, 2011), Chapter 14: "Framework Silicates", pp. 345–360.